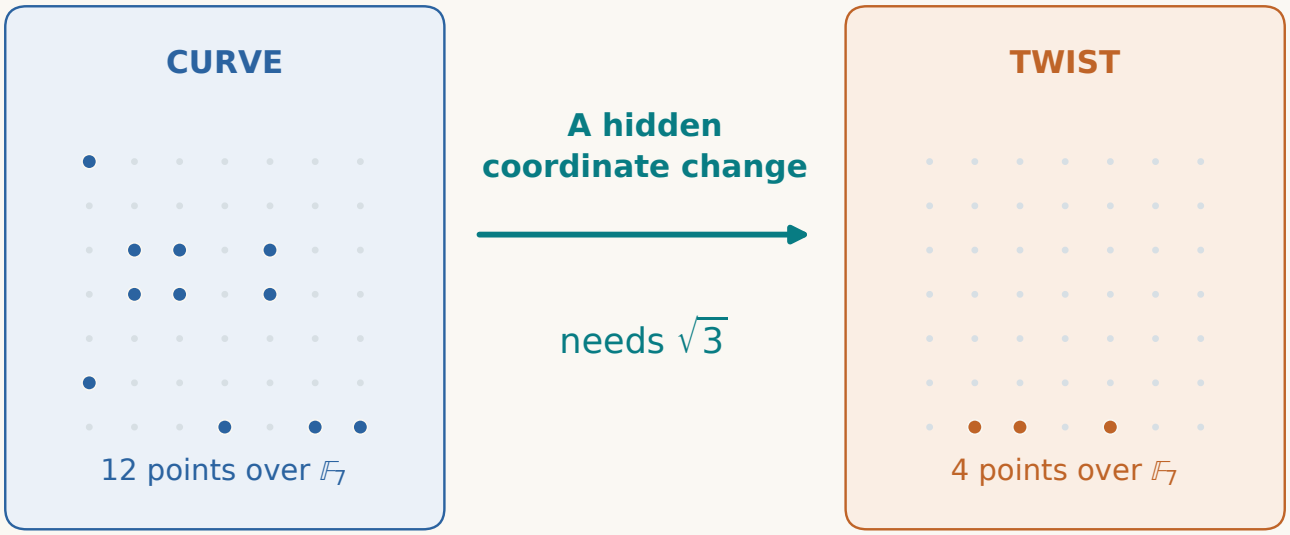
Same curve. Different visible points.

A visual introduction to quadratic, cubic, quartic and sextic twists — with finite-field examples and cryptographic intuition.



Add the missing root: both have 48 points over $\mathbb{F}_{49}$.

The big idea. A twist can look different when coordinates must come from a small field. After adding a missing root, an exact coordinate map reveals the same elliptic curve.

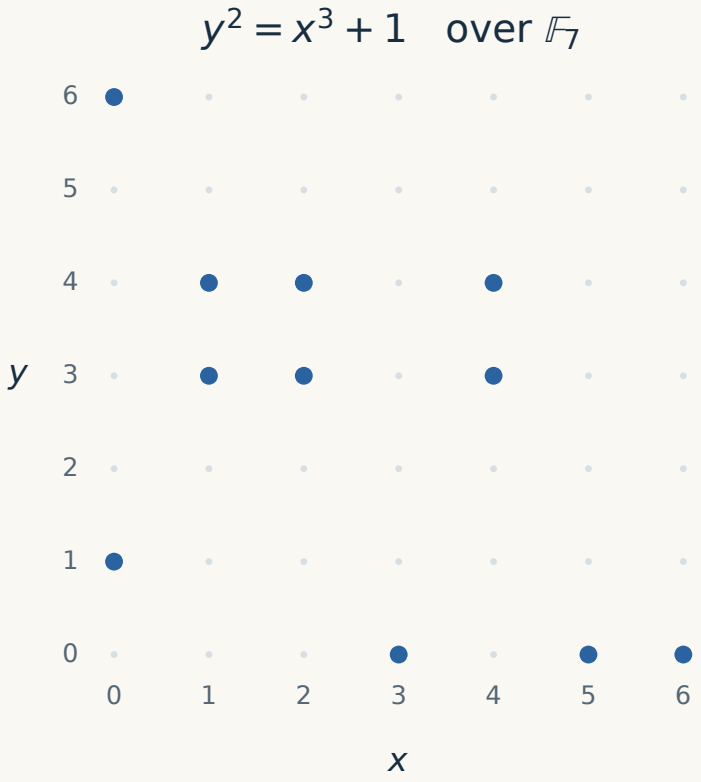
How to read this guide. Follow the pictures first. Then read the equations as instructions for the hidden map. All plotted finite-field points and point counts are exact.

REMEMBER

A twist changes how a curve is defined over your field, while preserving its geometry over an algebraic closure.

A curve is a rule for allowed points.

Over a finite field, an elliptic curve gives a finite set of coordinate pairs, together with a group law and an identity point O.



ONE VALID POINT

$(x, y) = (1, 3)$

$3^2 = 9 \equiv 2 \pmod 7$

11 plotted points

$+ O = 12$ in the group

Read the grid. The axes contain the residues 0 through 6. A blue dot marks a pair satisfying $y^2 = x^3 + 1$ modulo 7. The pale dots are rejected pairs.

Check one. At (1,3), the two sides are 9 and 2. They agree modulo 7. The point O is the identity of the group and is not shown in this affine grid.

REMEMBER

The point count includes O. The group law adds points on the curve; it is not ordinary coordinate-wise addition.

Your field is your coordinate toolbox.

A field tells you which numbers may appear in a coordinate or in the formula for a change of coordinates.

BASE FIELD $\mathbb{F}_7$

$0, 1, 2, 3, 4, 5, 6$

Squares: 0, 1, 2, 4

No s here has $s^2 = 3$.

→

EXTENSION $\mathbb{F}_{49}$

$a + bu, \quad u^2 = 3$

$a, b \in \mathbb{F}_7$

$7 \times 7 = 49$ elements

We add an algebraic tool, not extra decimal precision.

$u^2 = 3$ makes new coordinate changes possible.

$\mathbb{F}_{49} \neq \mathbb{Z}/49\mathbb{Z}$

A missing tool. In F_7 the square residues are 0, 1, 2 and 4. There is no square root of 3. A map needing that root cannot be written with F_7 coefficients.

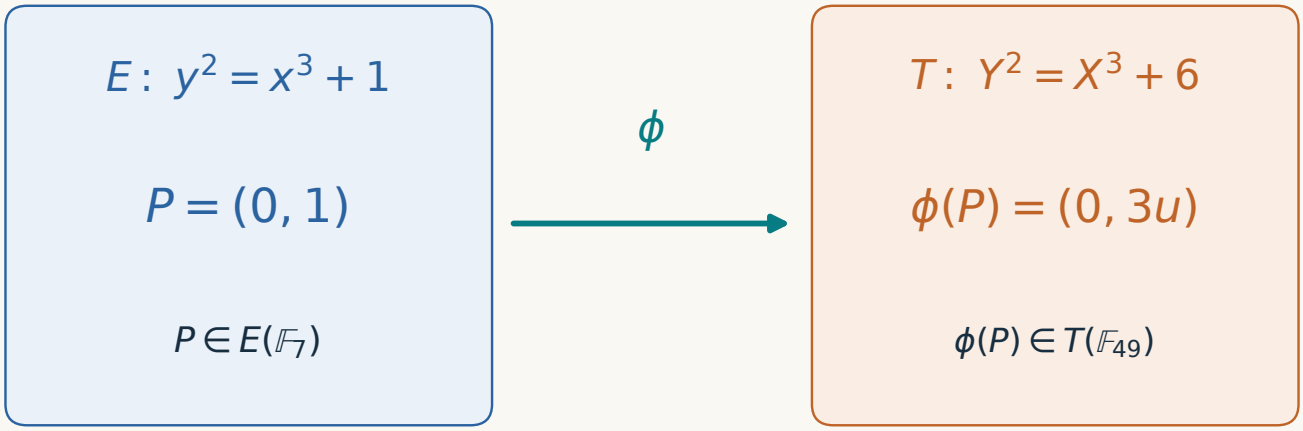
Add the tool. Introduce u with $u^2 = 3$. The numbers $a + bu$ form F_{49}. The original field sits inside it as the elements with $b = 0$.

REMEMBER

An extension adds algebraic numbers. F_{49} is a field of 49 elements, not arithmetic modulo the integer 49.

What a twist actually does.

These two equations describe isomorphic elliptic curves once the coordinate field contains u, a square root of 3.



$$\phi(x, y) = (3x,\ 3uy), \qquad u^2 = 3$$

$(3uy)^2 = 6y^2 = (3x)^3 + 6$ in characteristic 7

The map exists over the extension; it need not preserve base-field coordinates.

Follow a point. The point (0,1) on E maps to (0,3u) on T. That image is valid over F_{49}, but its y-coordinate is outside F_7.

Why counts differ. An isomorphism over the extension need not send the F_7-points of one model to the F_7-points of the other. Over F_{49} both groups have 48 points.

REMEMBER

Two curves are twists when they become isomorphic over an extension. A twist is trivial if the isomorphism already exists over the base field.

Flip the square / nonsquare choice.

Keep f(x) = x³ + 1. Compare y² = f(x) with 3v² = f(x) over F₇, where 3 is a nonsquare.

		$E: y^2 = f(x)$	$T: 3v^2 = f(x)$	
x	$f(x)$	number of y values	number of v values	
0	1	● ●	0	= 2
1	2	● ●	0	= 2
2	2	● ●	0	= 2
3	0	●	●	= 2
4	2	● ●	0	= 2
5	0	●	●	= 2
6	0	●	●	= 2

$$11 + 1 = 12 \quad + \quad 3 + 1 = 4 \quad = 16$$

Each group also has its own identity point O.

Each x balances. A nonzero square gives 2 points on the first model and 0 on the twist; a nonsquare gives 0 and 2. A zero gives 1 on each.

The point-count rule. Add the two identity points: #E + #T = 2(q + 1). Here 12 + 4 = 16. This twist model is F₇-isomorphic to Y² = X³ + 6.

REMEMBER

For a nonsquare quadratic twist: if #E = q + 1 − t, then #T = q + 1 + t.

The powers 2 and 3 do the work.

Start with $y^2 = x^3 + Ax + B$. Scaling x by u^2 and y by u^3 keeps the Weierstrass shape and scales A and B.

$$(x, y) \mapsto (u^2 x, u^3 y)$$
$$A \mapsto u^4 A \qquad B \mapsto u^6 B$$

A and B both nonzero	B = 0	A = 0
$u^4, u^6 \in K$	$u^4 \in K$	$u^6 \in K$
$u^2 \in K$	$u^4 = c$	$u^6 = c$
QUADRATIC	QUARTIC	SEXTIC

The coefficient powers explain the available root degrees.

$$u^3 = a \Rightarrow u^6 = a^2$$

Cubic twists live inside this family.

Generic curves. If A and B are nonzero, both u^4 and u^6 must be in the field. Their ratio is u^2, so a square root is all the freedom available.

Special curves. When a coefficient vanishes, one condition disappears. Fourth roots or sixth roots can then define new forms. The cubic case sits inside the sextic family.

REMEMBER

The degree describes the field needed for an isomorphism. The isomorphism is a one-to-one coordinate change, not a map of that degree.

Symmetry gives a map room to hide.

For characteristic greater than 3, these are the geometric symmetries that preserve the identity O. “Geometric” allows algebraic-closure coefficients.

$j \neq 0, 1728$



$(x, y) \mapsto (x, -y)$

2 geometric symmetries

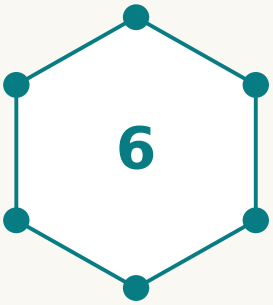
$j = 1728$



$(x, y) \mapsto (-x, iy)$

4 geometric symmetries

$j = 0$



$(x, y) \mapsto (\omega x, -y)$

6 geometric symmetries

$$i^2 = -1, \quad \omega^3 = 1, \quad \omega \neq 1$$

Polygons show symmetry groups, not the shape or points of the elliptic curve.

Why symmetry matters. Apply a field symmetry to the hidden map: you may get another map between the same curves. The discrepancy is a symmetry of the curve itself.

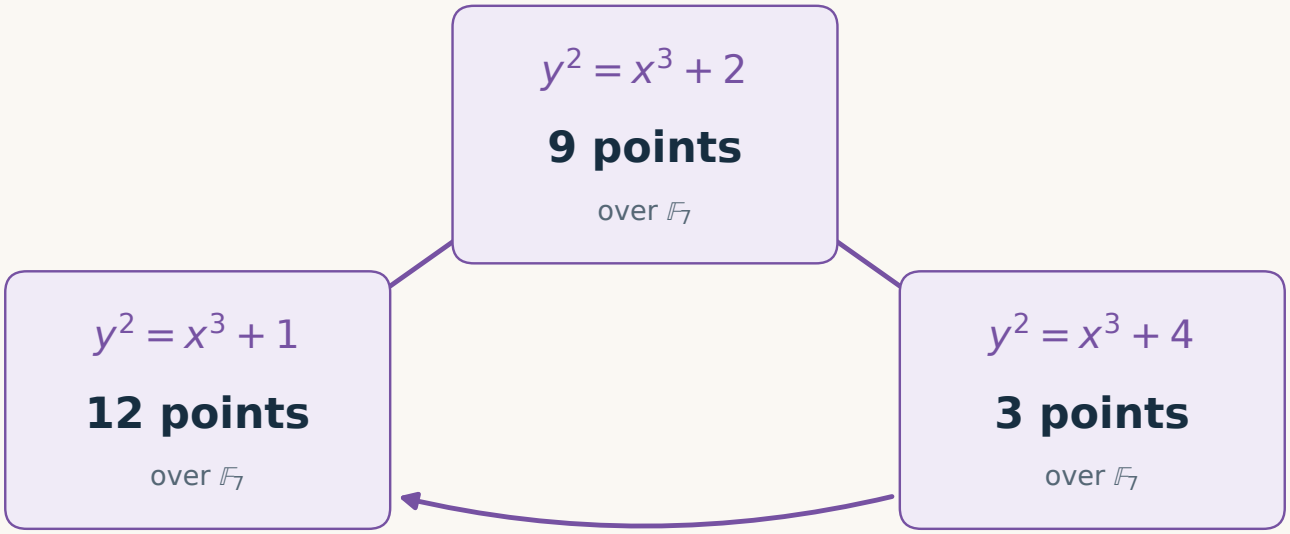
What j labels. The j-invariant is a number identifying the geometric curve. The special labels j = 1728 and j = 0 mark extra symmetry, which permits more twists.

REMEMBER

The same j-invariant means the same geometric elliptic curve. It does not determine the number or group structure of base-field points.

Cubic twists form a three-way family.

When A = 0, use $y^2 = x^3 + a^2B$. A cube root $v^3 = a$ removes the coefficient change.



$$B \mapsto a^2 B, \qquad v^3 = a$$

All three become isomorphic over $\mathbb{F}_{7^3}$.

The hidden map. From the twisted model back to the original, send (x,y) to (x/v², y/a). The cube root appears in the x-coordinate.

Notice the square. The multiplier on B is a², even though this is called a cubic twist. Over F_7 the square multipliers are 1, 2 and 4.

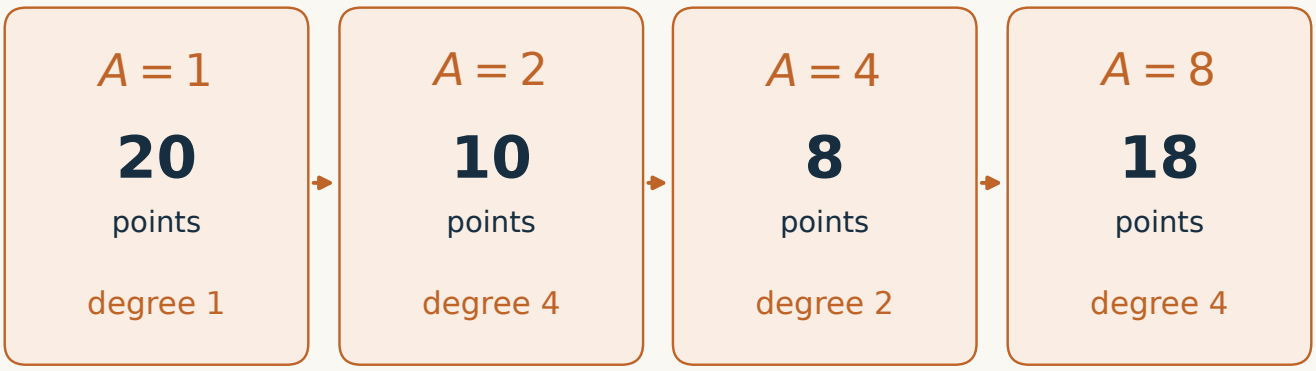
REMEMBER

Cubic refers to the missing cube root in the coordinate map. For q ≡ 1 mod 3, there are two nontrivial cubic twists.

Quartic twists need B to vanish.

For $y^2 = x^3 + Ax$, changing A to cA can be undone by a fourth root $u^4 = c$. The coefficients here are successive powers of 2 in F_{13}.

$$y^2 = x^3 + Ax \quad \text{over } \mathbb{F}_{13}$$



$$A \mapsto 2A \qquad u^4 = 2$$

Four forms: original, two exact quartic twists, one quadratic twist.

Read the four models. The coefficients are A = 1, 2, 4 and 8. The displayed degree is the smallest extension over which each becomes isomorphic to the A = 1 model.

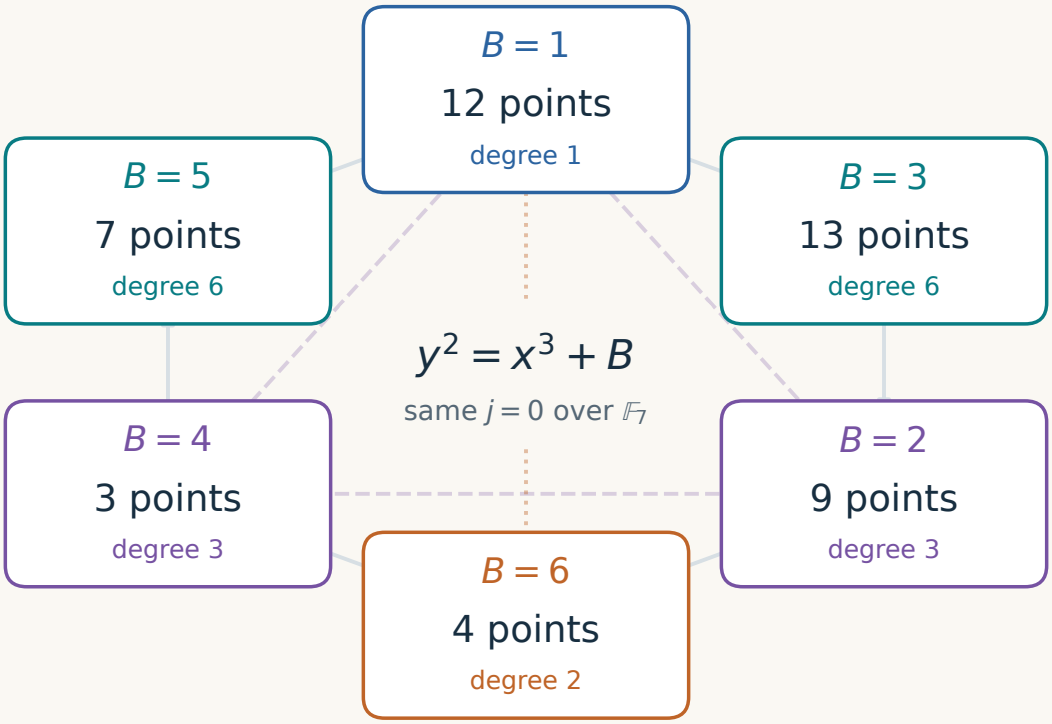
Find the quadratic member. A = 4 is a square multiplier. It is the quadratic twist of the original. A = 2 and A = 8 require degree 4.

REMEMBER

A quartic-twist family includes the original and its quadratic twist, as well as the members of exact degree 4.

One sextic family holds all six forms.

For $y^2 = x^3 + B$ over F_7, let $B = 3^i$ for $i = 0,\ldots,5$. Every node has $j = 0$; the point counts can be very different.



Blue: original Orange: quadratic Purple: cubic Teal: exact sextic

Read the colors. Relative to B = 1: B = 6 is quadratic; B = 2 and 4 are cubic; B = 3 and 5 are exact sextic twists. Degree 1 is the original.

Combine the moves. Multiplying B by a square gives a cubic-twist operation. Multiplying it by a cube gives a quadratic-twist operation. Together they generate this six-class family.

REMEMBER

“Sextic family” does not mean every member needs degree 6. The exact degrees here are 1, 2, 3, 3, 6 and 6.

Six possible symmetries ≠ six classes.

Two multipliers describe the same sextic class exactly when their ratio is a sixth power in the base field.

Over $\mathbb{F}_7$

$c^6 = 1$ for every $c \neq 0$

1	2	3
4	5	6

6 sextic classes

Over $\mathbb{F}_5$

$c^6 \in \{1, 4\}$

class 1	class 2
1, 4	2, 3

2 sextic classes

$$\#\text{classes} = \gcd(6, q - 1)$$

Count the classes. Fq* is cyclic of order q − 1. Modding out by n-th powers leaves gcd(n, q − 1) classes. Use n = 2, 4 or 6 for the relevant family.

A useful surprise. Over F_5 there are two j = 0 classes, not six. Both have 6 points. Counting points alone therefore cannot identify the twist class.

REMEMBER

Always name the base field. The j-invariant identifies the geometric curve; power classes distinguish its forms over that field.

Frobenius asks: are you visible here?

Frobenius raises each coordinate to the q-th power. Its fixed points are exactly the Fq-rational points.

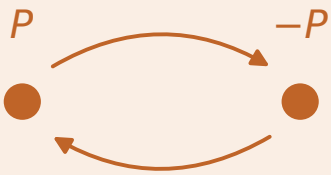
FIXED



$\pi(P) = P$

Coordinates lie in $\mathbb{F}_q$.

SWAPPED WITH ITS NEGATIVE



$\pi(P) = -P$

$\pi^2(P) = P$

$$\pi(x, y) = (x^q, y^q)$$

Under the twist isomorphism, twist-rational points satisfy the right-hand condition.

Original curve. A point with Fq coordinates stays fixed. After identifying a quadratic twist with the curve over Fq², its rational points satisfy π(P) = −P instead.

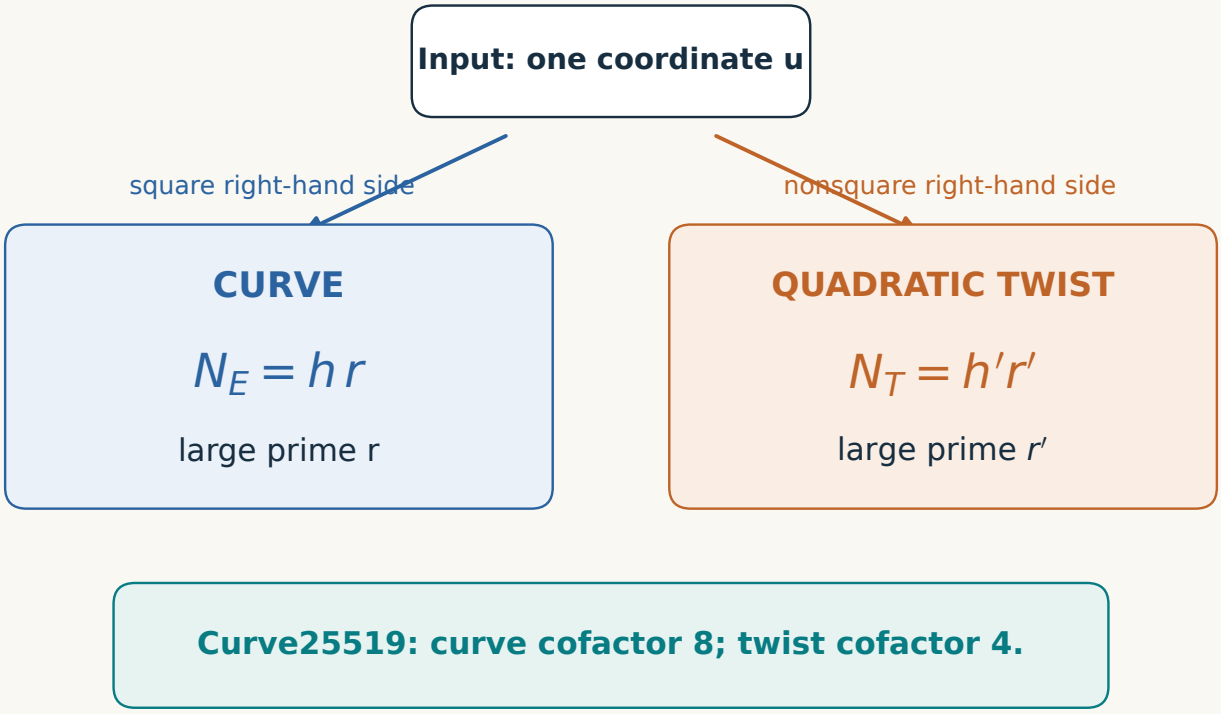
Two steps remove the sign. Applying Frobenius twice fixes an anti-fixed point. Quadratic twists become isomorphic over Fq². Points with P = −P can satisfy both conditions.

REMEMBER

Twisting changes the rule for which geometric points count as rational over the base field.

Some inputs can land on the twist.

For Montgomery single-coordinate scalar multiplication, a nonzero right-hand side selects the curve or its quadratic twist by its square class.



Why group orders matter. Small-order inputs can make scalar multiplication reveal information modulo a small factor. A large prime subgroup on the twist helps support the intended security.

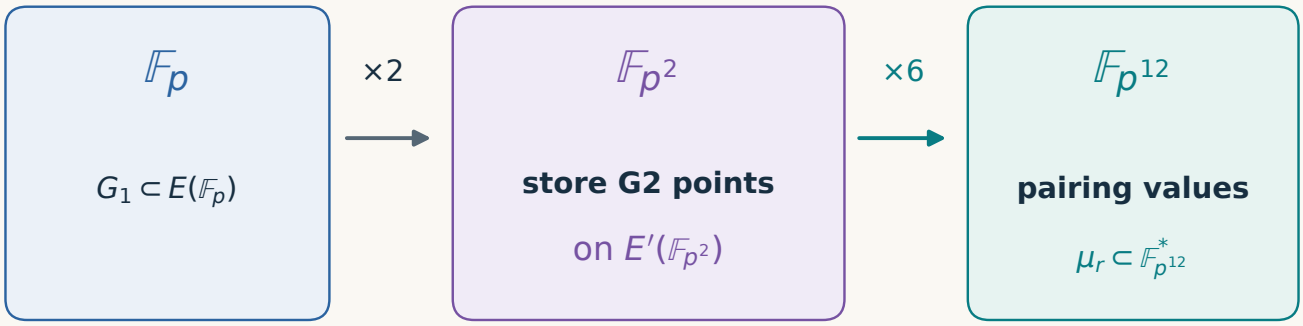
Follow the protocol. Twist security works alongside the required input, subgroup and all-zero-output rules. Unchecked full-coordinate inputs can reach additional curves too.

REMEMBER

Twist security means checking the arithmetic of the relevant twist group, not only that of the main curve.

A sextic twist makes G₂ cheaper.

BLS12-381 uses a special j = 0 curve. Its G₂ subgroup is represented on a sextic twist with coordinates in Fp².



$$E : y^2 = x^3 + 4 \qquad E' : y^2 = x^3 + 4(1+i)$$

$$w^6 = 1 + i, \qquad \phi(x, y) = (x/w^2, y/w^3)$$

The untwist maps the stored G2 subgroup into $E(\mathbb{F}_{p^{12}})$.

Embedding degree = 12 over Fp. Twist degree = 6 over Fp².

Smaller coordinates. Store and manipulate G₂ points on E′ over Fp². The untwist sends this subgroup to the corresponding points of E over Fp¹².

Keep the objects distinct. Elliptic-curve points live on E or E′. Pairing outputs live in a multiplicative subgroup of Fp¹². The twist does not move all of E(Fp¹²) into E′(Fp²).

REMEMBER

The twist degree 6 is measured over Fp². The embedding degree 12 is measured over Fp.

A compact map you can remember.

Assume characteristic greater than 3. Each formula below specifies a family; its exact degree can be smaller.

FAMILY	CURVE	COEFFICIENT CHANGE	MISSING TOOL
Quadratic	any j	$A \to d^2A$; $B \to d^3B$	square root of d
Cubic	j = 0	$B \to a^2B$	cube root of a
Quartic	j = 1728	$A \to cA$	fourth root of c
Sextic	j = 0	$B \to cB$	sixth root of c

Check your intuition

1. A curve over F_{101} has 96 points. How many on its nonsquare quadratic twist?
2. Does j = 0 guarantee six distinct twists over every finite field?
3. In F_7, why can the same geometric curve have groups of sizes 12 and 4?

Answers. 1: 108, because $2(101 + 1) - 96 = 108$. 2: No; the full sextic family has $\gcd(6, q - 1)$ classes. 3: The isomorphism needs extension-field coefficients and need not preserve F_7 coordinates.

REMEMBER

Ask: Which field? Which coefficient changes? Which root makes the map possible?

The boundaries of the picture.

The main guide assumes p > 3, nonsingular elliptic curves and isomorphisms that preserve the identity O.

Characteristics 2 and 3. At j = 0 the geometric symmetry groups become nonabelian, with orders 24 and 12 respectively. The class counts below include the original curve. In either characteristic, j ≠ 0 gives two classes.

BASE FIELD	CLASSES AT j = 0
F_{2^m}, m odd	3
F_{2^m}, m even	7
F_{3^m}, m odd	4
F_{3^m}, m even	6

Extensions of degree 8 can occur in characteristic 2. Quadratic binary twists use Artin–Schreier equations $s^2 + s = \delta$ rather than nonsquare parameters. The simple 2, 3, 4, 6 classification therefore needs the p > 3 assumption. [2]

An odd-characteristic exception. At j = 1728 with q ≡ 3 mod 4, the usual nonsquare quadratic-twist construction gives an isomorphic curve. A second form still exists in the quartic family. [2]

Sources and further study

[1] Steven Galbraith, Mathematics of Public Key Cryptography, chapter 9, §§9.4–9.5. Definitions, automorphisms and twists.

[2] Kronberg, Soomro and Top, Twists of Elliptic Curves (2017). Small characteristic, class counts and capitulation.

[3] RFC 7748, Elliptic Curves for Security. Montgomery ladders, Curve25519 and its twist.

[4] RFC 9380, §8.8 and Appendix G.3. BLS12-381 curve equations and Frobenius-based operations.

[5] Steven Galbraith, chapter 26, §26.6.2. Why twists improve pairing representations and arithmetic.

Verification. The F_7 and F_{13} point counts were enumerated directly; the displayed F_7 group structures and the two F_{49} counts were checked computationally. Diagrams are original. The separate image bundle contains PNG and vector SVG versions.